

Assignment 2: Computational Fluid Dynamics (ME543), Aug-Nov 2024

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- Provide: (i) the grid detail, (ii) the discretized equations detail, (iii) the boundary condition implementation detail, (iv) the required output (plots/any other such means), (v) a well-documented code.
- Items (i), (ii), and (iii) above should be written out/typed on a separate sheet and attached before items (iv). Soft copy of document containing items (i) to (iv) and code need to be submitted.
- Solve the following problems using finite difference method. If not provided, you can decide your own grid spacing and determine roughly the *coarsest* grid for *sufficient* accuracy, say 10^{-6} . If the analytical solution can be found for a problem then compare the numerical results with the analytical solution by plotting them together.

1. Consider a two-dimensional rectangular plate of dimension $L = 0.3$ m in the x -direction and $H = 0.4$ m in the y -direction. The material of the plate has a thermal conductivity of 380 W/m-K and a thermal diffusivity of 11.234×10^{-5} m²/s. Its edges are subjected to the following constant temperatures: (i) $y = 0$, $T = 40$ °C, (ii) $x = 0$, $T = 0$ °C, (iii) $y = H$, $T = 10$ °C, and (iv) $x = L$, $T = 0$ °C. The steady-state temperature distribution in the plate is to be determined. Initially choose a uniform grid size of 0.01 m in both directions (later you can change the grid sizes). Solve the algebraic equations using (i) point Gauss-Seidel method with successive overrelaxation (SOR), (ii) line Gauss-Seidel method with SOR, and (iii) ADI method. For each iterative method, take initial guess for temperature to be equal to zero.

The analytical solution for the steady-state temperature distribution in this case is given by

$$T(x, y) = T_{y=0}f(x, H - y) + T_{y=H}f(x, y)$$

where

$$f(x, y) = 2 \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n\pi} \frac{\sinh\left(\frac{n\pi y}{L}\right)}{\sinh\left(\frac{n\pi H}{L}\right)} \sin\left(\frac{n\pi x}{L}\right)$$

- (a) Show the 2-norm of residual of the algebraic system of equation versus iteration for all methods together on a plot (keep $\omega = 1$).
- (b) Use the relaxation factor in the range ($0.6 \leq \omega \leq 1.4$), in steps of 0.2, and report the number of iterations required for convergence. Note that you should have the same initial guess and the convergence criterion in each case.

2. Consider the plate of problem 1. The transient temperature distribution in the plate is to be determined. The plate is initially at a uniform temperature of 0 °C. The boundary conditions are kept the same. Solve the problem using (a) fully explicit method, (b) fully implicit method, (c) Crank-Nicolson method and (d) ADI method. Use a suitable method for solving the algebraic equations coming from implicit methods.

- (a) Compute the solution till steady-state is reached. You will need to determine and implement a criterion for the determination of the “steady-state”. Compare it with the analytical solution.
- (b) Plot the time history of the temperatures at plate center obtained from all the methods.
- (c) Do you get the same steady-state solution for large time steps? Explain.

3. Solve the wave equation $u_t + u_x = 0$ on a domain $0 \leq x \leq 40$ for the following two initial conditions

$$(i) u(x, 0) = \sin \frac{2n\pi x}{40}, \quad (ii) u(x, 0) = \begin{cases} 1 & x \leq 10 \\ 0 & x > 10 \end{cases}$$

with periodic boundary conditions. Choose mesh with $\Delta x = 1$ and 0.5 and compute to $t = 18$. Take $n = 1$ and 3 for the first initial condition. Take Courant number $C = 1, 0.6, 0.3$ and compare graphically with the exact solution. Use following schemes for both the initial conditions: (a) first-order upwind, (b) Lax, (c) MacCormack and (d) second-order upwind. Comment on the results.

Coding guidelines/tips

- Refer assignment 1.
- You can compare 2D contours by plotting them together as follows

